30 Variance of Estimators

- The bias function of an estimator measures the estimator's tendency, over many hypothetical samples, to overestimate or underestimate the parameter, as a function of potential values of the parameter.
- But bias is only one consideration. Bias only considers the average value of the estimator over many samples, which is just one feature of the estimator's sample-to-sample distribution.

Example 30.1. Consider again estimating μ for a Poisson(μ) distribution based on a random sample $X_1, \dots, X_n$ of size n. We have seen that both $\bar{X}$ and S^2 are unbiased estimators of μ. If we want to choose between these two estimators, how do we decide?

1. Assume $n = 3$ and $\mu = 2$. Describe in full detail how you could conduct a simulation to approximate the sample-to-sample distribution of $\bar{X}$ and its and its expected value and standard deviation. Then conduct the simulation and record the results. What does the standard deviation measure?

2. Repeat part 1 for S^2.

3. Compare the simulation results for $\bar{X}$ and S^2 when $n = 3$ and $\mu = 2$. Based on the simulation results, which estimator of μ is preferred when $\mu = 2$: $\bar{X}$ or S^2? Why? But then explain why this information by itself isn't very helpful.

- A statistic (or estimator) is a characteristic of the *sample* which can be computed from the data. More precisely, a statistic is a function of $X_1, \dots, X_n$, but not of θ.
- Because the sample is random, a statistic is itself a *random variable*, and therefore has its own probability distribution, which describes how values of the statistic would vary from sample-to-sample over many (hypothetical) samples.
- Statistics exhibit **sample-to-sample variability**: the value of a statistic varies from sample to sample.
  - For many statistics (e.g., means and proportions)— but not all— statistics from larger random samples vary less, from sample to sample, than statistics from smaller random samples.
  - For example, $\mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$.
- When choosing between two *unbiased* estimators, the one with smaller variance is generally preferred.
- Remember that the variance of an estimator will be a function of the unknown parameter θ, so we need to consider the variance *function* for various potential values of θ.

Example 30.2. Consider again estimating μ for a Poisson(μ) distribution based on a random sample $X_1, \dots, X_n$ of size n. We have seen that both $\bar{X}$ and S^2 are unbiased estimators of μ. If we want to choose between these two estimators, how do we decide?

1. Identify the variance function of $\bar{X}$.

2. Describe in full detail how you could use simulation to approximate the variance function of S^2.

3. It can be shown that

$$\mathrm{Var}(S^2) = \frac{2\mu^2}{n-1} + \frac{\mu}{n}, \qquad \text{when the population distribution is Poisson}(\mu)$$

   Sketch a plot of the variance functions of both $\bar{X}$ and S^2.

4. Which of these two unbiased estimators of μ, $\bar{X}$ or S^2, is preferred? Why?

5. Suppose $n = 3$ and the sample is $(3, 0, 2)$. For this sample $\bar{x} = 1.67$ and $s^2 = 2.33$. Which number, 1.67 or 2.33, is a better estimate of μ? Explain.

6. Suppose $n = 3$ and the sample is $(3, 0, 2)$. For this sample $\bar{x} = 1.67$ and $s^2 = 2.33$. Which number, 1.67 or 2.33, would you choose as the estimate of μ based on this sample? Why?